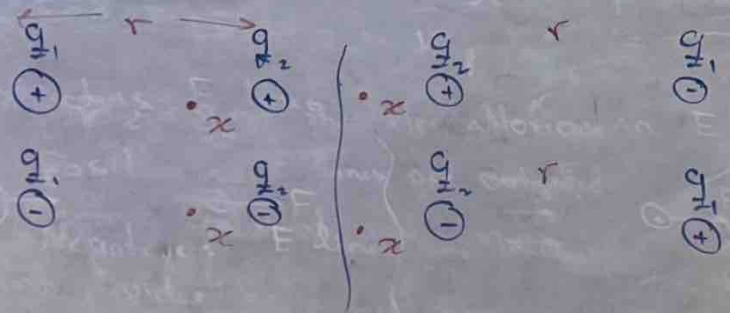


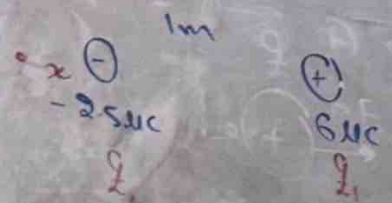
Physics II



$$x = \frac{r}{\sqrt{\frac{q_1}{q_2}} + 1}$$

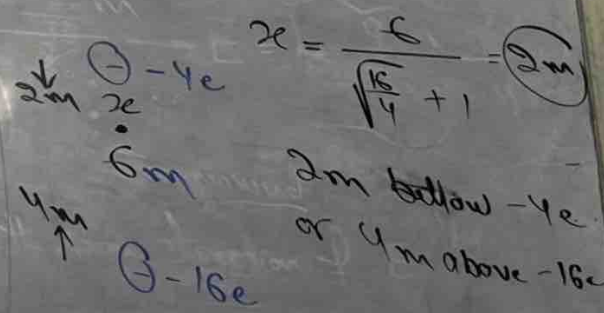
Same signs
Opposite signs

where is the electric force should be zero.



$$x = \frac{1}{\sqrt{\frac{6}{2.5}} - 1} = 1.82m \text{ (left of } -2.5\mu C)$$

$$= 2.82m \text{ (left of } 6\mu C)$$



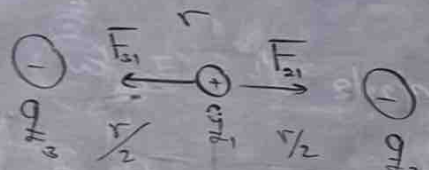
$$x = \frac{6}{\sqrt{\frac{16}{4}} + 1} = 2m$$

2m below -4e
or 4m above -16e

Physics II

Q111 if $F = k_e \frac{k^2}{r^2}$, $q_1 = e$, $q_2 = -e$, $q_3 = -2e$

Find the electric force on q_1 placed at middle.



$$F_{21} = k_e \frac{q_2 q_1}{r_{21}^2} = k_e \frac{e \cdot e}{(r/2)^2} = k_e \frac{e^2}{r^2} = 4 \left(k_e \frac{e^2}{r^2} \right) \rightarrow 4F$$

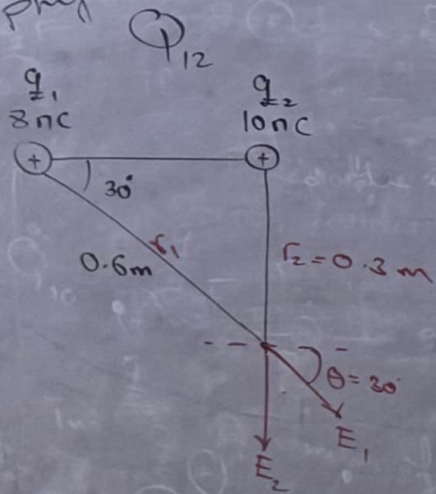
$$F_{31} = k_e \frac{q_3 q_1}{r_{31}^2} = k_e \frac{2e \cdot e}{(r/2)^2} = k_e \frac{8e^2}{r^2} = 8 \left(k_e \frac{e^2}{r^2} \right) \rightarrow 8F$$

$$8F \leftarrow \text{---} \rightarrow 4F$$

$$F_r = 4F - 8F = -4F$$

$$F_r = 4F \text{ at } \theta = 180^\circ$$

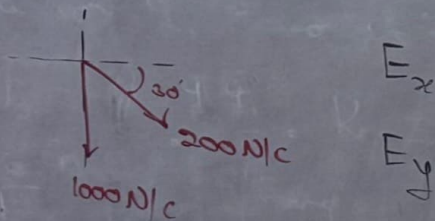
Physics II



$$\therefore \sin 30 = \frac{r_2}{0.6} \rightarrow r_2 = 0.6 \sin 30 = 0.3 \text{ m}$$

$$E_1 = k_e \frac{q_1}{r_1^2} = 9 \times 10^9 \times \frac{8 \times 10^{-9}}{0.6^2} = 200 \text{ N/C}$$

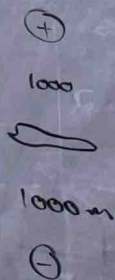
$$E_2 = k_e \frac{q_2}{r_2^2} = 9 \times 10^9 \times \frac{10 \times 10^{-9}}{0.3^2} = 1000 \text{ N/C}$$



Physics II

Air plane at 2000m from ground, +40C at 3000 from ground, -40C at 1000m from ground

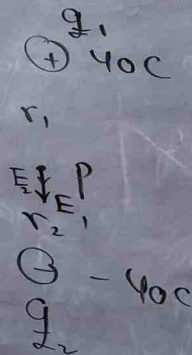
Find the electric field at air plane



Find the electric for
 $\oplus +40 \mu\text{C}$
 $\ominus -40 \mu\text{C}$

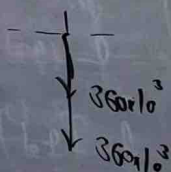
$$F = k \frac{q_1 q_2}{r_{12}^2}$$

$$F = 9 \times 10^9 \times \frac{40 \times 40}{2000^2}$$



$$E_1 = k \frac{q_1}{r_1^2} = 9 \times 10^9 \times \frac{40}{1000^2} = 360 \times 10^3 \text{ N/C}$$

$$E_2 = k \frac{q_2}{r_2^2} = 9 \times 10^9 \times \frac{40}{1000^2} = 360 \times 10^3 \text{ N/C}$$



$$E = -360 \times 10^3 - 360 \times 10^3 = -720 \times 10^3 \text{ N/C}$$

$$E = 720 \times 10^3 \text{ N/C at } \theta = 270^\circ$$